Durgesh Kumar

About

Hello, I am Durgesh. I study category theory and quantum foundations.

I recently completed my master's in mathematics at the University of Calgary, supervised by Robin Cockett. My master's thesis explored a generalisation of the correspondence between orthonormal bases and copyable elements of Frobenius algebras to the infinite-dimensional setting. Before that, I studied mathematics and physics at Ashoka University. In winter 2026, I visited the Compositional Systems and Methods and Logic and Semantics groups at TalTech.

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Education

University of Calgary Jan 2024 – Apr 2026
M.Sc. in Mathematics (thesis-based); Calgary, Canada
CGPA: 4.00/4.00. Supervisor: Robin Cockett.

Ashoka University Aug 2021 – May 2022
Postgraduate Diploma in Research; Sonipat, India
Advanced major in mathematics; minor in physics.
GPA: 3.76; magna cum laude.

Ashoka University Aug 2018 – May 2021
B.Sc. in Mathematics; Sonipat, India
Concentration in physics.

Research experience

Iliad Intensive Deferred to a later London cohort
LISA, London
Iliad Intensive is a competitive, funded, month-long programme hosted at LISA for mathematicians interested in AI safety.

Visiting Researcher Winter 2026
Tallinn University of Technology (TalTech); Tallinn, Estonia
Compositional Systems and Methods and Logic and Semantics groups.

ACT Adjoint School participant Feb - Aug 2023
University of Maryland, College Park; United States
Project: Message Passing Logic for Categorical Quantum Mechanics.
Supervisor: Priyaa Varshinee Srinivasan (NIST).

Research Assistant Jun 2022 - Dec 2023
Ashoka University; India
Project: Monoidal Categories and Computation.
Supervisor: Debayan Gupta.

Publications

Unitary, Inner Product, and Dagger Categories 2026
Robin Cockett, Durgesh Kumar, and Priyaa Varshinee Srinivasan.
Applied Category Theory (ACT) 2026.
Accepted to the 9th International Conference on Applied Category Theory, Tallinn, Estonia, July 2026.

Master's thesis

An adjunction between finiteness spaces and algebras in Lefschetz spaces 2026
Durgesh Kumar. Master's thesis in mathematics, University of Calgary.
Supervisor: Robin Cockett.
Nominated by the thesis committee for a national award.

Research in progress

We are working on generalising the notion of a non-stationary dagger developed in our paper to chiral settings. Our main aim is to connect it to the literature on linearly distributive categories and thus to categorical semantics of quantum mechanics. Further work will explore applications of this framework to modelling concurrent quantum protocols.

Teaching experience

Graduate Teaching Assistant, University of Calgary

MATH 271	Summer 2025
MATH 271, MATH 311	Winter 2025
MATH 271	Fall 2024
MATH 267, MATH 211	Winter 2024

Outreach and exposition

Linear Actegories

September 2023

Paige Frederick, Durgesh Kumar, and Fabian Wiesner.
n-Category Café. We wrote an expository blog post on linear actegories as part of the ACT Adjoint School.

Applied Category Theory

2022

Ashoka University. I designed and co-taught a study module in applied category theory.
Course materials

For my blogs on formalisation and AI safety, see Mathematics and AI projects.

Talks, seminars, and posters

An adjunction between finiteness spaces and algebras in Lefschetz spaces — March 2026
Talk. TallCat Seminar; TalTech, Tallinn, Estonia.

Comparative study of different formalisms of contextuality — Fall 2025
Semester-long seminar. Quantum Group seminar series; University of Calgary.

Sheaf-theoretic view of contextuality — September 2025
Talk. Peripatetic Seminar; University of Calgary.

Categorical semantics of infinite-dimensional quantum theory — August 2025
Poster. International Quantum Horizons Symposium; Banff. | Poster (PDF)

A study of Lefschetz spaces — June 2025
Talk. FMCS 2025; University of Ottawa. | Slides (PDF)

Towards Frobenius algebras in infinite-dimensional quantum mechanics — May 2025
Talk. Alberta Mathematical Dialogues; University of Calgary. | Slides (PDF)

Towards Frobenius algebras in infinite-dimensional quantum mechanics — February 2025
Talk. Peripatetic Seminar; University of Calgary.

The category of Lagrangian relations — July 2024
Talk. FMCS 2024; Kananaskis. | Slides (PDF)

Honors and awards

Master's thesis nomination — 2026
Nominated by the thesis committee for a national award.

Mitacs Globalink Research Award - Quantum Stream — 2026
Tallinn University of Technology.

Mitacs Globalink Research Award - Quantum Stream 2025
University of Edinburgh; withdrawn because ATAS clearance was not received in time.

Dean's List 2020, 2022
Ashoka University; Monsoon 2020 and Spring 2022.

Ignite National Award for Grassroot Innovations 2012
Department of Science and Technology, Government of India.

Mathematics and AI projects

Heisenberg meets Anju

Notion of agents in quantum foundations and artificial intelligence
This post explores how the notion of agents in extended Wigner's friend scenarios in quantum foundations can add nuance to agentic frameworks in AI safety.

In programmes we trust: A categorical perspective

A (very) short sketch of the Curry-Howard-Lambek isomorphism
This post explores the trustworthiness of Lean certifications of AI-assisted (and AI-driven!) mathematics by giving a categorical account of the Curry–Howard–Lambek correspondence.

Inverse Galois Problem

We (Team Dirac) participated in the Inverse Galois Problem Competition, organised by the SAIR Foundation, and finished in 14th place.
Leaderboard | Code

what's my sheaf?

A global section calculator for empirical models based on Abramsky and Brandenburger's sheaf-theoretic formalism of contextuality.

Ongoing work

We (Team Dirac) are currently working on the Andrews–Curtis Conjecture (ACC) Challenge, organised by the SAIR Foundation, and hovering in the top 10 across the competition categories (leaderboard). We also have an ongoing project on formalising some results in mathematical general relativity in Lean (repository), and a project exploring some computational evidence in the Indus script (repository).

Patent

System to Integrate Driving License Validation in Vehicular Control
Patent record: 3408/DEL/2012.

Relevant coursework

Measure Theory; Algebraic Topology; Functional Analysis; Elliptic Curves; Algebraic Number Theory; Fourier Series; General Relativity and Cosmology; Mathematical Logic; Topics in Quantum Computation and Quantum Information; Category Theory.